

Queries

```
USE healthcare;  
CREATE TABLE patient_flow AS  
SELECT *  
FROM healthcare_analytics_patient_flow_data;
```

```
SELECT *  
FROM patient_flow  
LIMIT 5;
```

```
SELECT COUNT(*) AS total_patients  
FROM patient_flow;
```

-- Which patient group experiences the longest wait time ?

-- Checking for gender.

```
SELECT AVG(`Patient Waittime`) AS avg_patient_time, `Patient Gender`  
FROM patient_flow  
GROUP BY `Patient Gender`;
```

-- AVG wait time is similar but we are having an inconsistent gender called femalemale.

```
SELECT DISTINCT `Patient Gender`, `Patient Id`  
FROM patient_flow;
```

```
SELECT `Patient Id`, `Patient Gender`  
FROM patient_flow  
WHERE `Patient Gender` != 'Male' AND `Patient Gender` != 'Female';
```

-- we will be clearing out the femalemale problem here by

```
SELECT AVG(`Patient Waittime`) AS avg_patient_time, `Patient Gender`  
FROM patient_flow  
WHERE `Patient Gender` = 'Male' OR `Patient Gender` = 'Female'  
GROUP BY `Patient Gender`;
```

-- by race

```
SELECT DISTINCT `Patient Race`  
FROM patient_flow;
```

```
SELECT AVG(`Patient Waittime`) AS avg_patient_time, `Patient Race`  
FROM patient_flow  
GROUP BY `Patient Race`;
```

-- everything is around 34 - 35 min. So no

-- age group.

```
SELECT AVG(`Patient Waittime`) AS avg_patient_time,
```

CASE WHEN

```
    `Patient Age` < 18 THEN 'Under 18'
  WHEN `Patient Age` >= 18 AND `Patient Age` < 30 THEN '18 - 30'
  ELSE '30 and above'
END AS age_group
FROM patient_flow
GROUP BY age_group;
-- Doesnt make any difference.
```

-- Q2 Which departments receive the highest patient volume?"

```
SELECT `Patient Id`, `Department Referral`, `Patient Admission Flag`
FROM patient_flow
LIMIT 15;
```

-- Total patients per dept. and of total admissions.

```
SELECT COUNT(`Patient Id`) AS total_count, `Department Referral`
FROM patient_flow
GROUP BY `Department Referral`;
```

-- So orthopedics had the most visits. 5400 people didn't visit any dept. so why did they come.
Maybe they missed coming in

-- Let's see how many people were admitted too.

```
SELECT `Department Referral`, COUNT(*) AS total_count,
      COUNT(`Patient Admission Flag`) AS admission_flag
FROM patient_flow
WHERE `Patient Admission Flag` = 'Admission'
GROUP BY `Department Referral`
ORDER BY total_count;
```

```
SELECT DISTINCT `Patient Admission Flag`
FROM patient_flow;
```

```
SELECT
  `Patient Admission Flag`,
  COUNT(*)
FROM patient_flow
GROUP BY `Patient Admission Flag`;
```

```
SELECT
  `Department Referral`,
  SUM(CASE WHEN `Patient Admission Flag` = 'Admission' THEN 1 ELSE 0 END) AS
  Admission_Count,
  SUM(CASE WHEN `Patient Admission Flag` = 'Not Admission' THEN 1 ELSE 0 END) AS
  Not_Admission_Count
FROM patient_flow
GROUP BY `Department Referral`;
```

```
SELECT DISTINCT `Patient Satisfaction Score`  
FROM patient_flow;
```

```
SELECT  
    AVG(`Patient Waittime`) AS avg_patient_time,  
    CASE  
        WHEN `Patient Satisfaction Score` > 0 AND `Patient Satisfaction Score` <= 4 THEN 'Not  
satisfied'  
        WHEN `Patient Satisfaction Score` = 5 THEN 'Moderately satisfied'  
        WHEN `Patient Satisfaction Score` > 5 AND `Patient Satisfaction Score` <= 10 THEN '  
Highly_Satisfied'  
    END AS Satisfaction_level  
FROM patient_flow  
GROUP BY `satisfaction_level`;
```